Mineralogy

- **What is mineral??**

A mineral is a naturally occurring homogeneous solid, inorganically formed, with a definite chemical composition and an ordered atomic arrangement.

or

A mineral is a body produced by the processes of inorganic nature, having usually a definite chemical composition and, if formed under favorable conditions, a certain characteristic atomic structure which is expressed in its crystalline form and other physical properties

Quartz

Physical properties of mineral

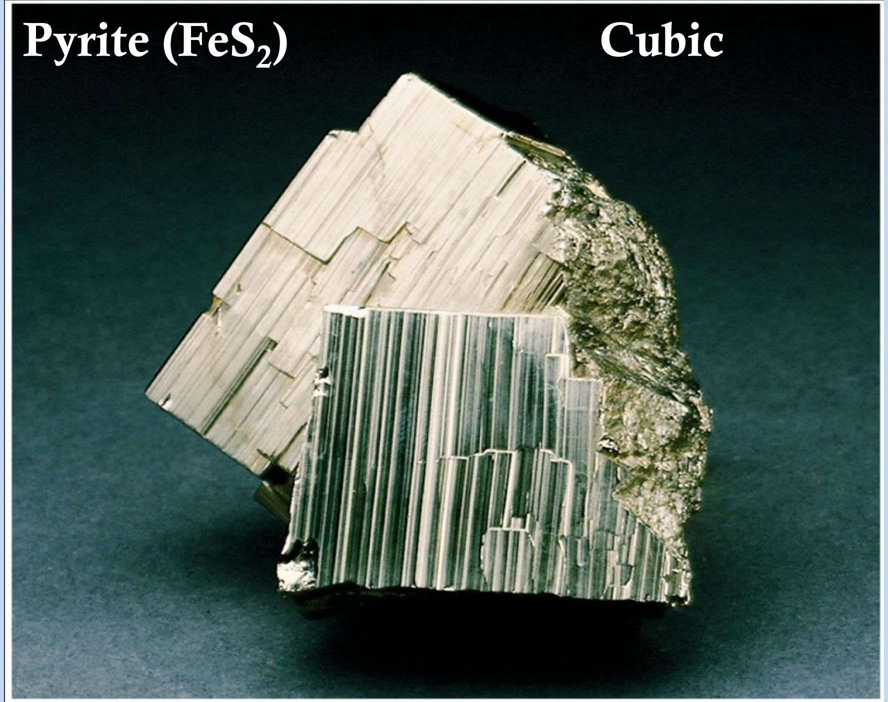
Minerals have distinguishing physical properties that in most cases can be used to determine the identity of the mineral.

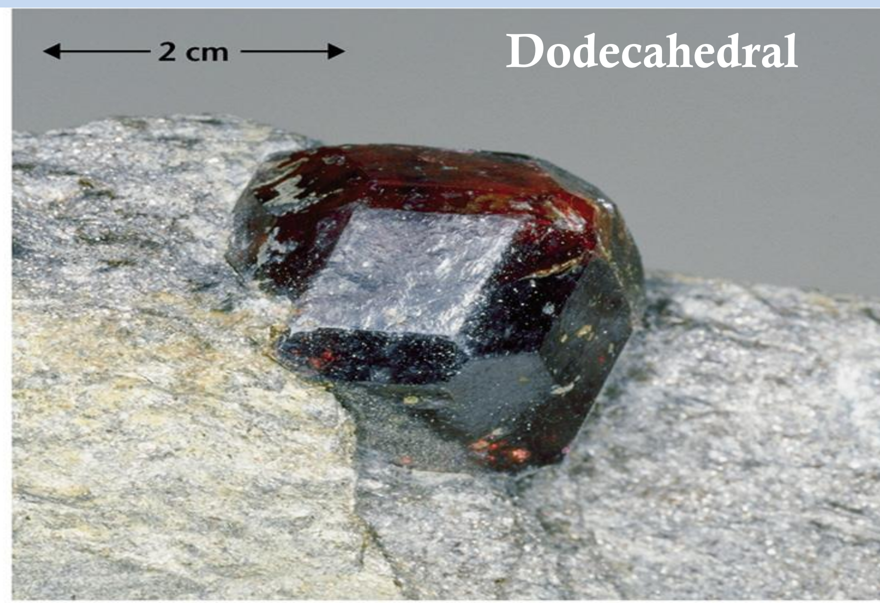
1. **Crystal Habit**
2. **Cleavage**
3. **Parting**
4. **Fracture**
5. **Hardness**
6. **Tenacity**
7. **Density (Specific Gravity)**
8. **Color**
9. **Streak**
10. **Luster**
11. **Play of Colors**
12. **Fluorescence and Phosphorescence**
13. **Magnetism**
14. **Other Properties**

1. Crystal Habit

The term used to describe general shape of a crystal is ***habit.*** In nature perfect crystals are rare. The faces that develop on a crystal depend on the space available for the crystals to grow. Crystals sometimes develop certain forms more commonly than others, although the symmetry may not be readily apparent from these common forms.

Habit	Common Example (s)
Acicular	Natrolite, Rutile
Bladed	Actinolite, Kyanite
Columnar	Calcite, Gypsum/Selenite
Cubic	Pyrite, Galena, Halite
Dendritic or arborescent	Pyrolusite and other Mn-oxide minerals, Magnesite, native copper
Fibrous	Serpentine group, Tremolite (i.e. Asbestos)
Foliated or micaceous or lamellar (layered)	Mica (Muscovite, Biotite, etc.)
Massive or compact	Limonite, Turquoise, Cinnabar, Realgar
Nodular or tuberose	Chalcedony, various Geodes
Octahedral	Diamond, Magnetite
Prismatic	Tourmaline, Beryl
Platy	Wulfenite


Pyrite (FeS_2)
Cubic


2 cm
Dodecahedral

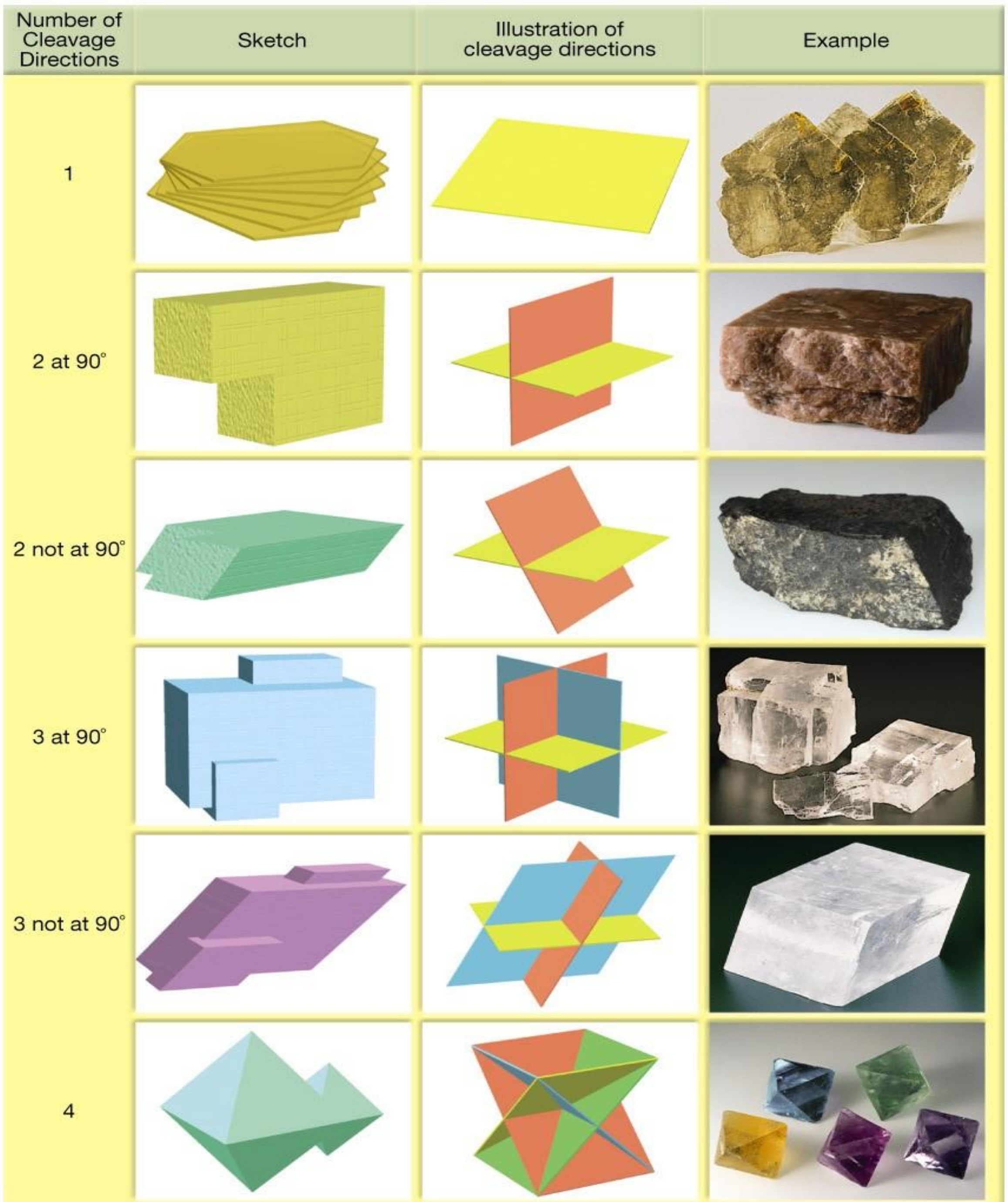
2. Cleavage

Crystals often contain planes of atoms along which the bonding between the atoms is weaker than along other planes. Cleavage occurs along planes in the crystal lattice. In such a case, if the mineral is struck with a hard object, it will tend to break along these planes. This property of breaking along specific planes is termed cleavage.

The cleavage can also be described in terms of its quality, i.e., if it cleaves along perfect planes it is said to be perfect, and if it cleaves along poorly defined planes it is said to be poor.

Cleavage can also be described by general forms names, for example if the mineral breaks into rectangular shaped pieces it is said to have **cubic cleavage (3 cleavage directions, Galena**), if it breaks into prismatic shapes, it is said to have **prismatic cleavage (2 cleavage directions**, **Tremolite**), or if it breaks along **basal pinacoids (1 cleavage direction**, **Mica**) it is said to have pinacoidal cleavage.

Cleavage can be described in the same manner that crystal forms are described.
For example if a mineral has cleavage along {100} it will break easily along planes parallel to the (100) crystal face, and any other planes that are related to it by symmetry.

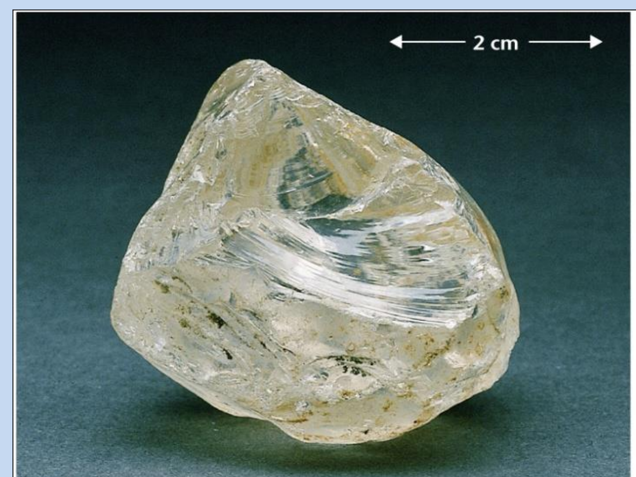


Cleavage

- splits along parallel planes of weakness
- not always present

Fracture

Random breakage of mineral
eg. Conchoidal fracture of **quartz**



3. Parting

Parting is also a plane of weakness in the crystal structure, but it is along planes that are weakened by some applied force. It therefore may not be apparent in all specimens of the same mineral, but may appear if the mineral has been subjected to the right stress conditions.

4. Fracture

If the mineral contains no planes of weakness, it will break along random directions called fracture. Several different kinds of fracture patterns are observed.

Conchoidal fracture - breaks along smooth curved surfaces **(Quartz).**

Fibrous and splintery - similar to the way wood breaks **(Kyanite).**

Hackly - jagged fractures with sharp edges **(Native Copper).**

Uneven or Irregular - rough irregular surfaces (**Pyrite, Magnetite**).

5. Hardness

Hardness is determined by scratching the mineral with a mineral or substance of known hardness. Hardness is a relative scale, thus to determine a mineral's hardness, one need materials with a hardness greater than the mineral and also need to know minerals of lesser hardness.

Hardness is determined on the basis of Moh's relative scale of hardness exhibited by some common minerals. These minerals are listed below, along with the hardness of some common objects.

Absolute hardness of a mineral determined by **Sclerometer**

Hardness	Mineral	Common Objects	Absolute Hardness
1	Talc		1
2	Gypsum	Fingernail (2+)	3
3	Calcite	Copper Penny (3+)	9
4	Fluorite		21
5	Apatite	Steel knife blade (5+), Window glass (5.5)	48
6	Orthoclase	Steel file	72
7	Quartz		100
8	Topaz		200
9	Corundum		400
10	Diamond		1500

6. Tenacity

Tenacity is the resistance of a mineral to breaking, crushing, or bending. Tenacity can be described by the following terms

Brittle - Breaks or powders easily.

Malleable **-** can be hammered into thin sheets.

Sectile **-** can be cut into thin shavings with a knife.

Ductile - bends easily and does not return to its original shape.

Flexible - bends somewhat and does not return to its original shape.

Elastic **-** bends but spring back after bending.

Density (Specific Gravity)

Density refers to the mass per unit volume. Specific Gravity is the relative density. In cgs units density is grams per cm^3, and since water has a density of 1 g/cm^3, specific gravity would have the same numerical value has density, but no units. Specific gravity is often a very diagnostic property for those minerals that have high specific gravities.

Mineral	Composition	Atomic # of Cation	Specific Gravity
Aragonite	$CaCO_3$	40.08	2.94
Strontianite	$SrCO_3$	87.82	3.78
Witherite	$BaCO_3$	137.34	4.31
Cerussite	$PbCO_3$	207.19	6.58

Mineral	Composition	Specific Gravity
Graphite	C	2.23
Quartz	SiO_2	2.65
Feldspars	$(K,Na)AlSi_3O_8$	2.6 - 2.75
Topaz	$Al_2SiO_4(F,OH)_2$	3.53
Corundum	Al_2O_3	4.02
Barite	$BaSO_4$	4.45
Pyrite	FeS_2	5.02
Galena	PbS	7.5

7. Color
Color is sometimes an extremely diagnostic property of a mineral, for example olivine and epidote are almost always green in color. But, for some minerals it is not at all diagnostic because minerals can take on a variety of colors. These minerals are said to be allochromatic. For example quartz can be clear, white, black, pink, blue, or purple.

8. Streak
Streak is the color produced by a fine powder of the mineral when scratched on a streak plate. Often it is different than the color of the mineral in non- powdered form

9. Luster
Luster refers to the general appearance of a mineral surface to reflected light. Two general types of luster are designated as follows:

Metallic - looks shiny like a metal. Usually opaque and gives black or dark colored streak.

Non-metallic - Non metallic lusters are referred to as

- ***vitreous*** - looks glassy - examples: clear quartz, tourmaline
- ***resinous*** - looks resinous - examples: sphalerite, sulfur
- **pearly** - iridescent pearl-like - example: apophyllite.
- ***greasy*** - appears to be covered with a thin layer of oil - example: nepheline.
- ***silky*** - looks fibrous. - examples - some gypsum, serpentine, malachite
- ***adamantine*** - brilliant luster like diamond

<u>Color</u>

Color <u>not </u>unique to each mineral

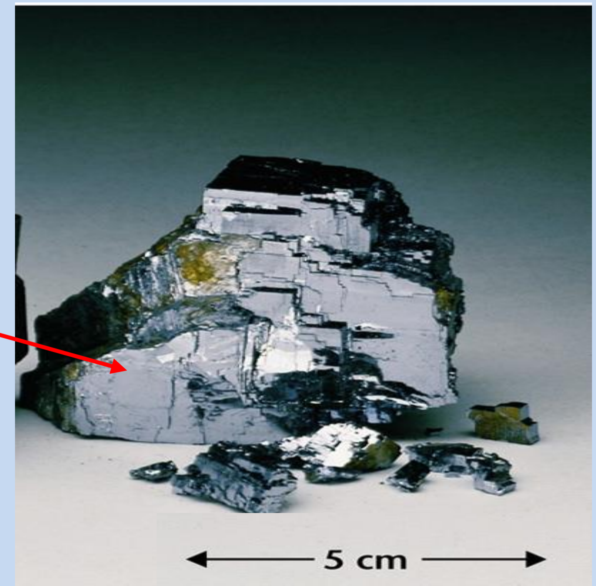
Eg. Quartz (SiO2) can be white, glassy, smoky & purple (amethyst)

<u>Streak</u>

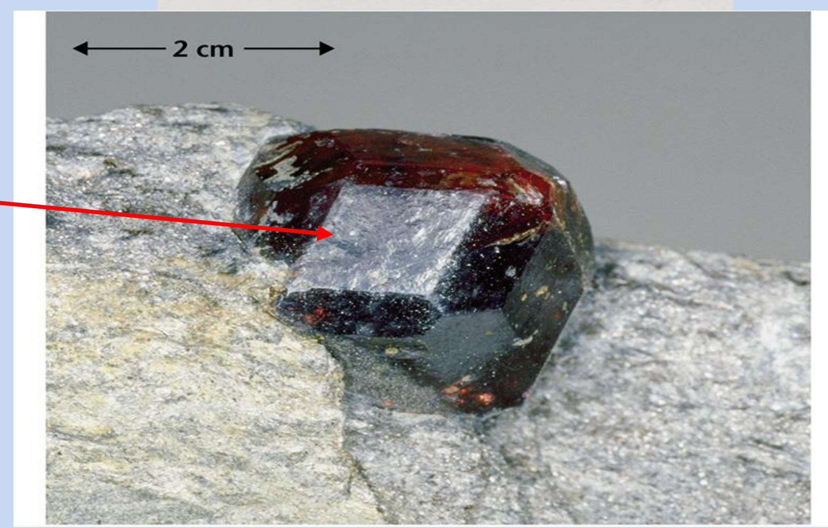
Streak is the color of powder when mineral scratched along porcelain plate.
Unique to each mineral → good for I.D.

Luster

- How light reflects off surface of mineral



- e.g. Galena (PbS) has metallic luster



- e.g. Garnet has vitreous luster

10. Fluorescence and Phosphorescence

Minerals that light up when exposed to ultraviolet light, x-rays, or cathode rays are called fluorescent. If the emission of light continues after the light is cut off, they are said to be phosphorescent.
Some specimens of the same mineral show fluorescence while other don't. For example some crystals of fluorite (CaF_2) , scheelite ($CaWO_4$), willemite (Zn_2SiO_4), calcite ($CaCO_3$), scapolite ($3NaAlSi_3O8 \cdot (NaCl - CaCO_3$), and diamond (C).

11. Magnetism

Magnetic minerals result from properties that are specific to a number of elements. Elements like Ti, Cr, V, Mn, Fe, Co, Ni, and Cu can sometimes result in magnetism. If any minerals contain these elements and show magnetic property are called ***paramagnetic*** minerals.

Minerals that do not have these elements, and thus have no magnetism are called ***diamagnetic***. Examples of diamagnetic minerals are quartz, plagioclase, calcite, and apatite.
Paramagnetic minerals only show magnetic properties when subjected to an external magnetic field. When the magnetic field is removed, the minerals have no magnetism.

Ferromagnetic minerals have permanent magnetism if the temperature is below the ***Curie Temperature***. These materials will become magnetized when placed in a magnetic field, and will remain magnetic after the external field is removed. Examples of such minerals are magnetite, hematite-ilmenite solid solutions (Fe_2O_3 - $FeTiO_3$), and pyrrhotite ($Fe_{1-x}S$).

Common classification of minerals and their basis

Minerals may be classified according to chemical composition. They are here categorized by anion group. The list below is in approximate order of their abundance in the Earth's crust.

a) Silicate class: (most rocks are ≥95% silicates) :

Examples: feldspars, quartz, olivines, pyroxenes, amphiboles, garnets, and micas.

b) Carbonate class: The carbonate class also includes the nitrate and borate minerals.

c) Sulfate class: Anhydrite , celestine, barite and gypsum

d) Halide class: The halide class includes the fluoride, chloride, bromide and iodide minerals.

e) Oxide class: Common oxides include hematite, magnetite, chromite

f) Sulfide class: Common sulfides pyrite , chalcopyrite and galena

g) Phosphate class: The phosphate class includes the phosphate, arsenate, vanadate

h) Element class: gold, silver, copper, semi-metals and non metals (antimony, graphite, sulfer)

i)Organic class: include various oxalates, citrates, acetates, hydrocarbons

CHEMICAL BONDING AND COMPOUND FORMATION

- Chemical bonds make atoms more stable than they are if non-bonded
- Bond formation involves changes in the electrons on two atoms

  This is achieved by one of two methods

  - Electron transfer
  - Electron sharing

Electron transfer involves creation of **ions**, which bond via **ionic** bonds to form ionic compounds. A Familiar compound like table salt, sodium chloride, is a classic example of an ionic compound.

Electron sharing involves the sharing of electrons between two atoms and the creation of **covalent** bonds.

Covalently bonded compounds typically have very different properties from ionic compounds, and they also involve combinations of different types of elements.

Sulfides: Dense, Usually Metallic Many Major Ores

- Major Cause of Acid Rain

4. Minerals can be identified by their physical properties = atomic structure

Carbonates

- Principal Components of limestone and dolostone
- Storehouse for CO_2
- Major regulator of climate

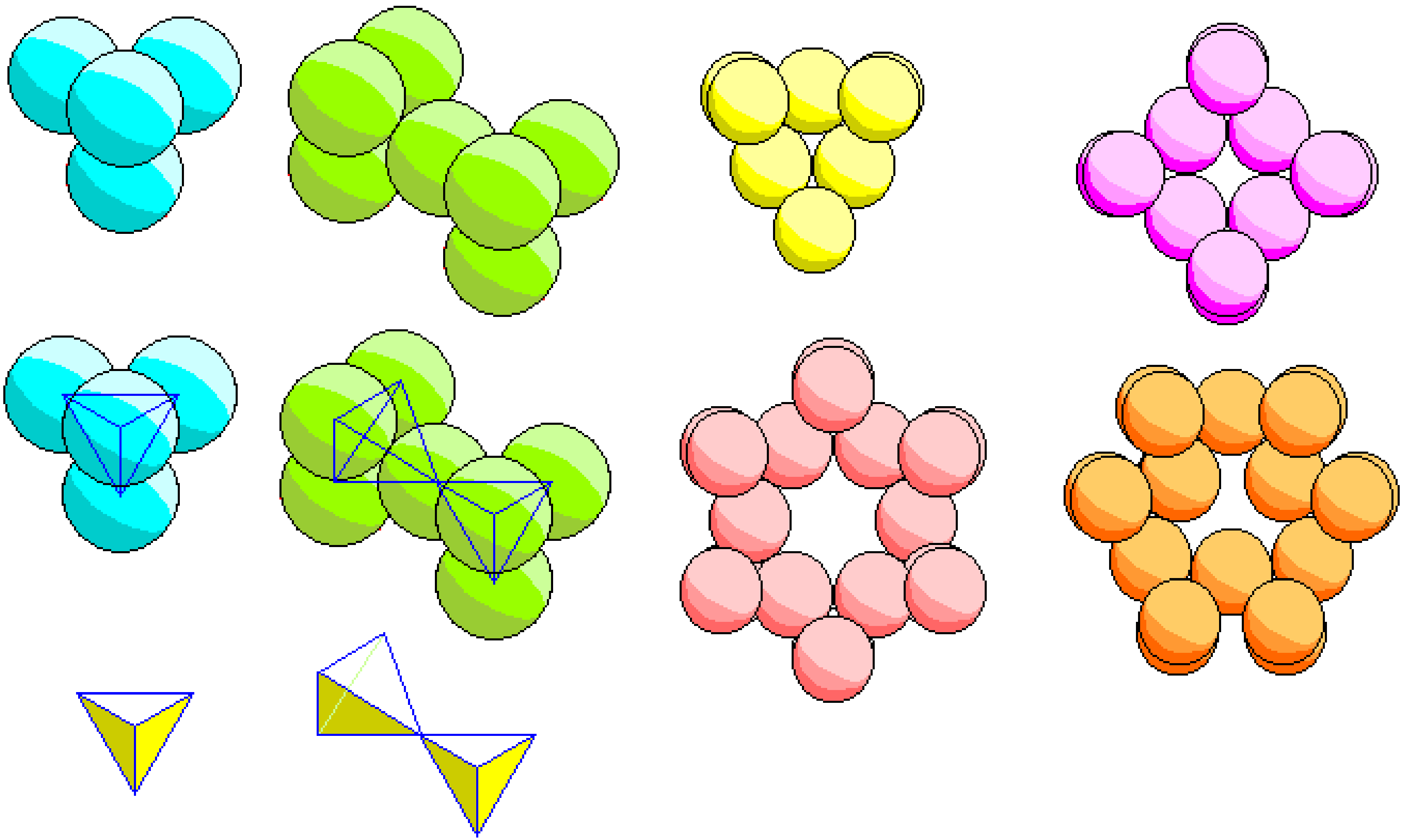
Oxide: Hematite

4. Minerals can be identified by their
physical properties = atomic structure

MOST IMPORTANT MINERAL SUITE:
The Silicate Minerals

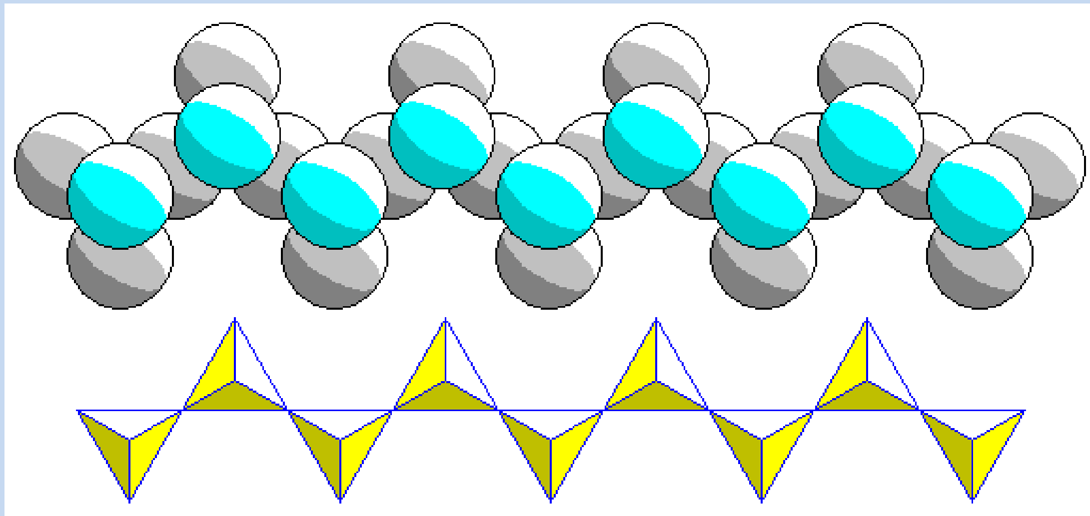
- Si + O = 75% of Crust
- Silicates make up 95% + of all Rocks
- SiO_4: -4 charge
- Link Corner-To-Corner by Sharing Oxygen atoms

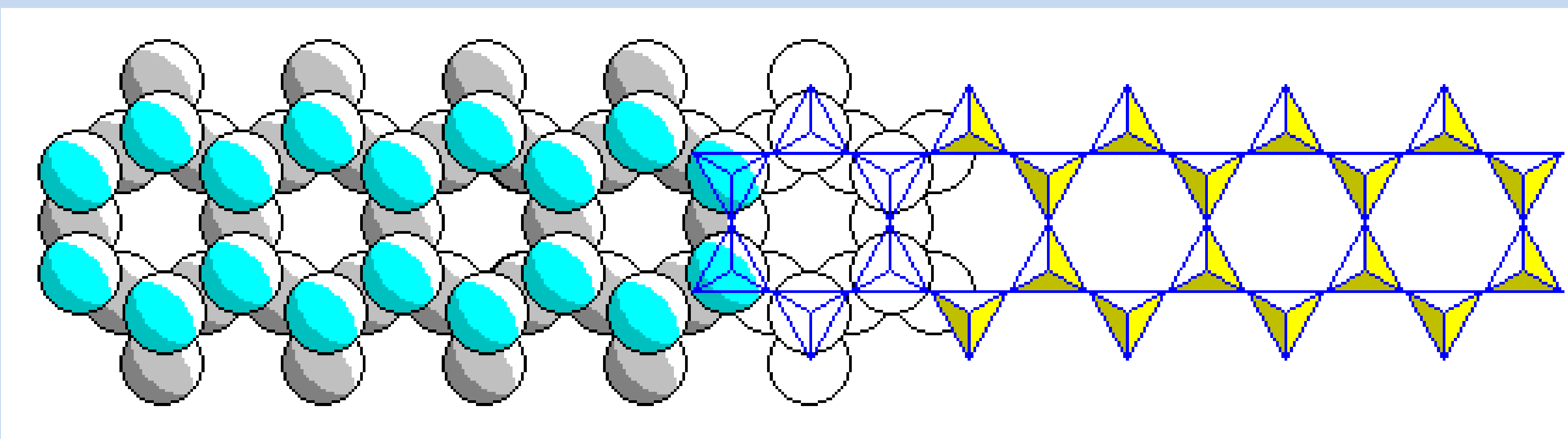
Silicate Structures



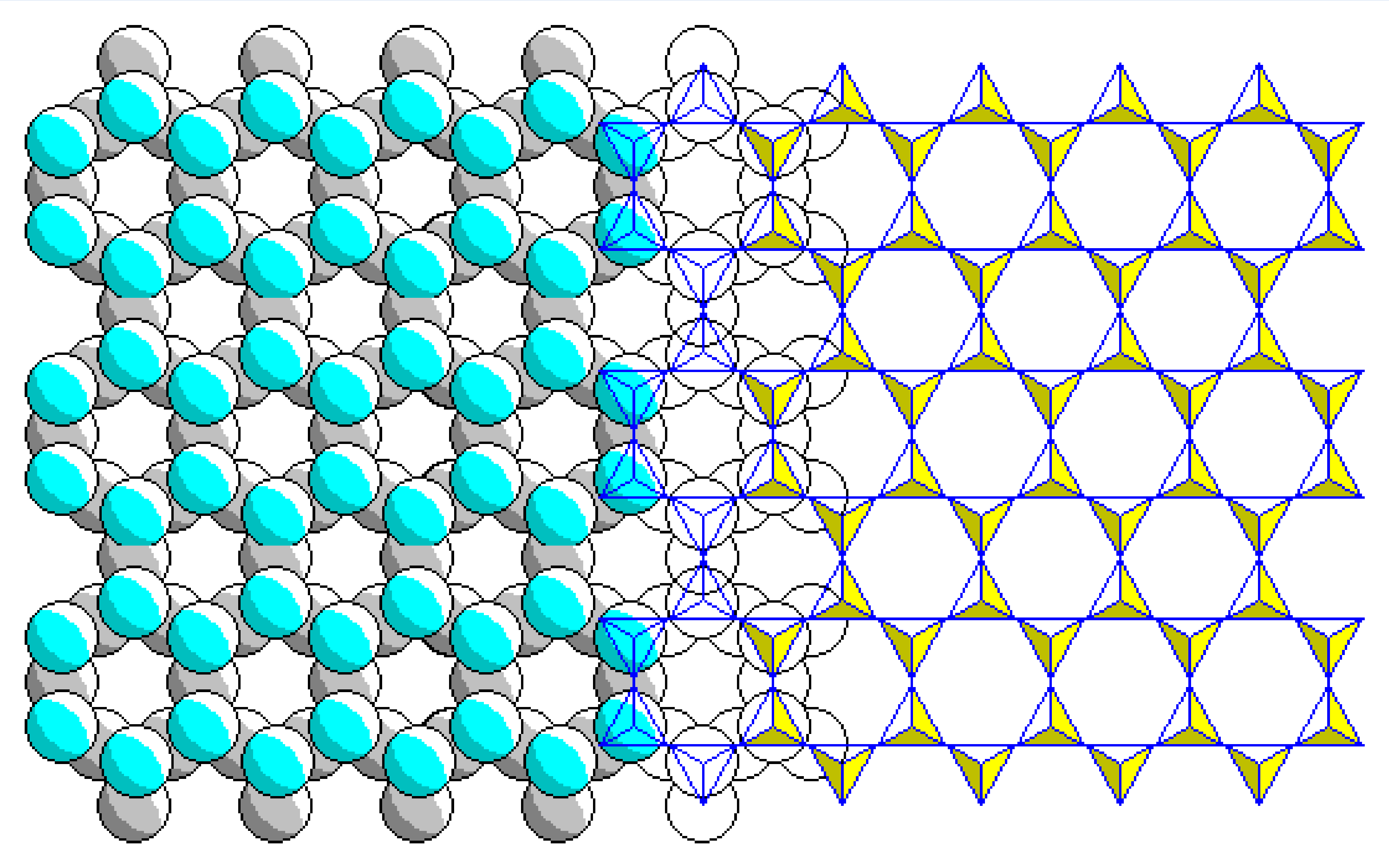
5. Silicates are the most important mineral group

Chain Silicates

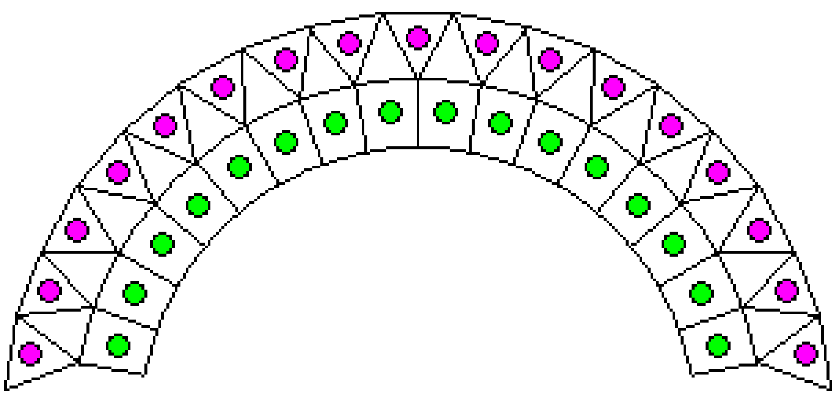


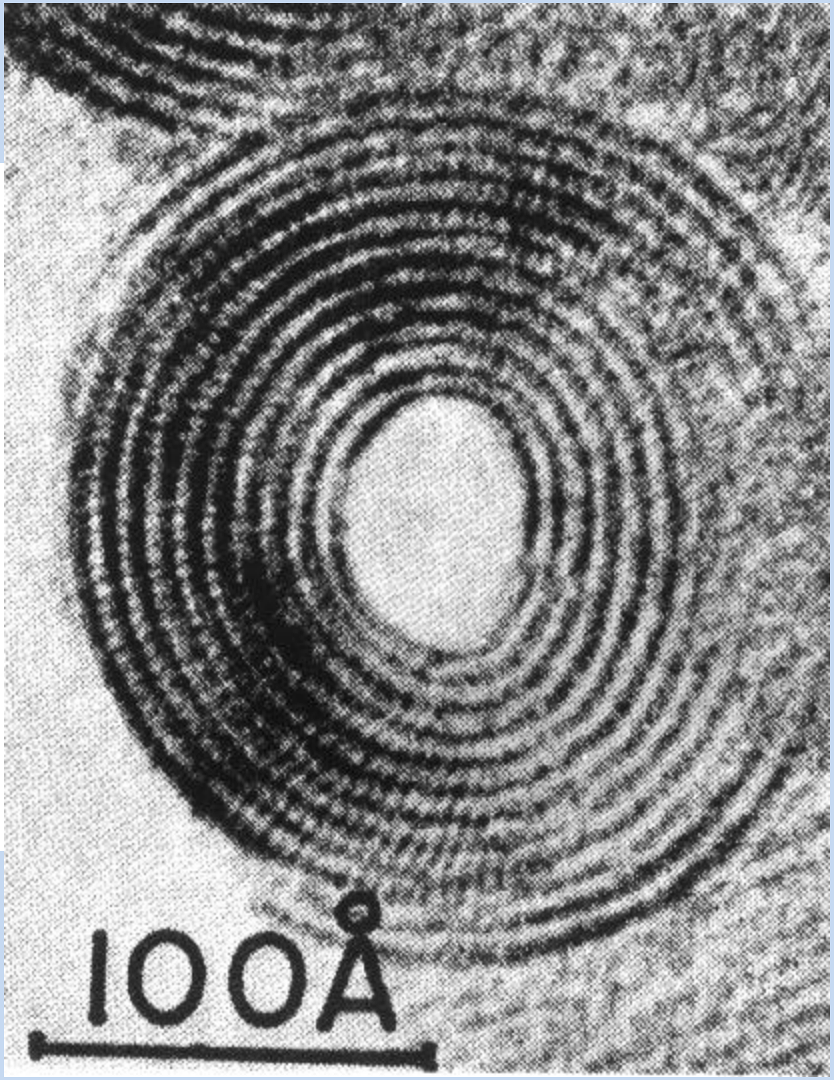


Sheet Silicates



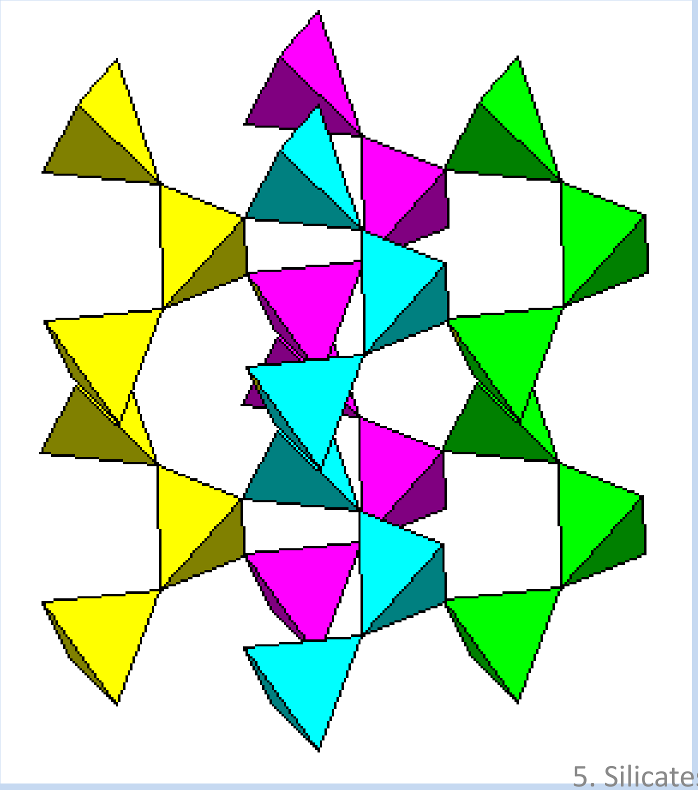
One Type of Asbestos



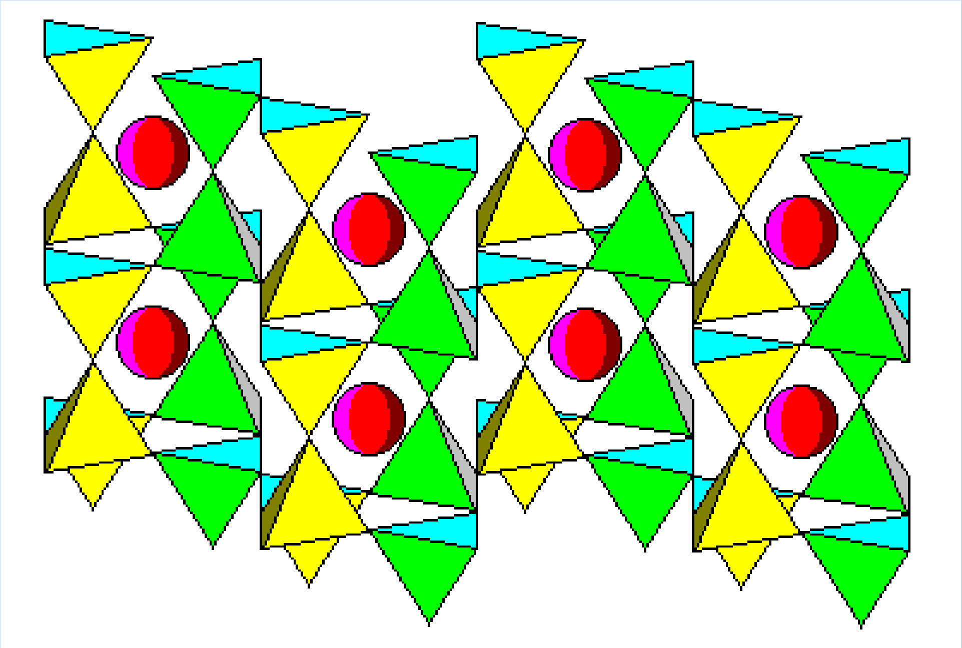


Tectosilicates - Three-Dimensional Networks

- Quartz

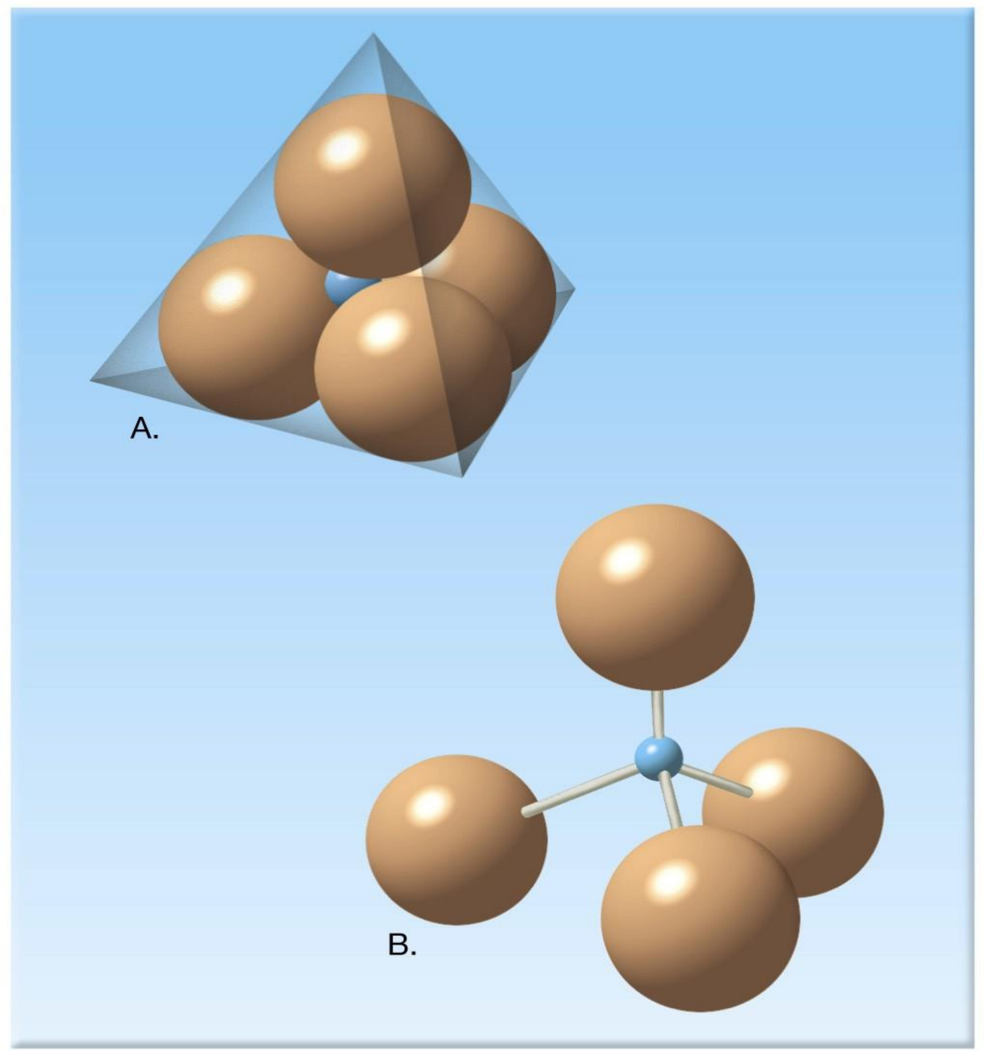


Feldspars



Mineral Groups - Silicates

- Most common type of mineral in crust
- Si-O tetrahedra
  - Fundamental building block
  - 4 O around 1 Si in 3-d pyramidal structure



Silicate Mineral Groups

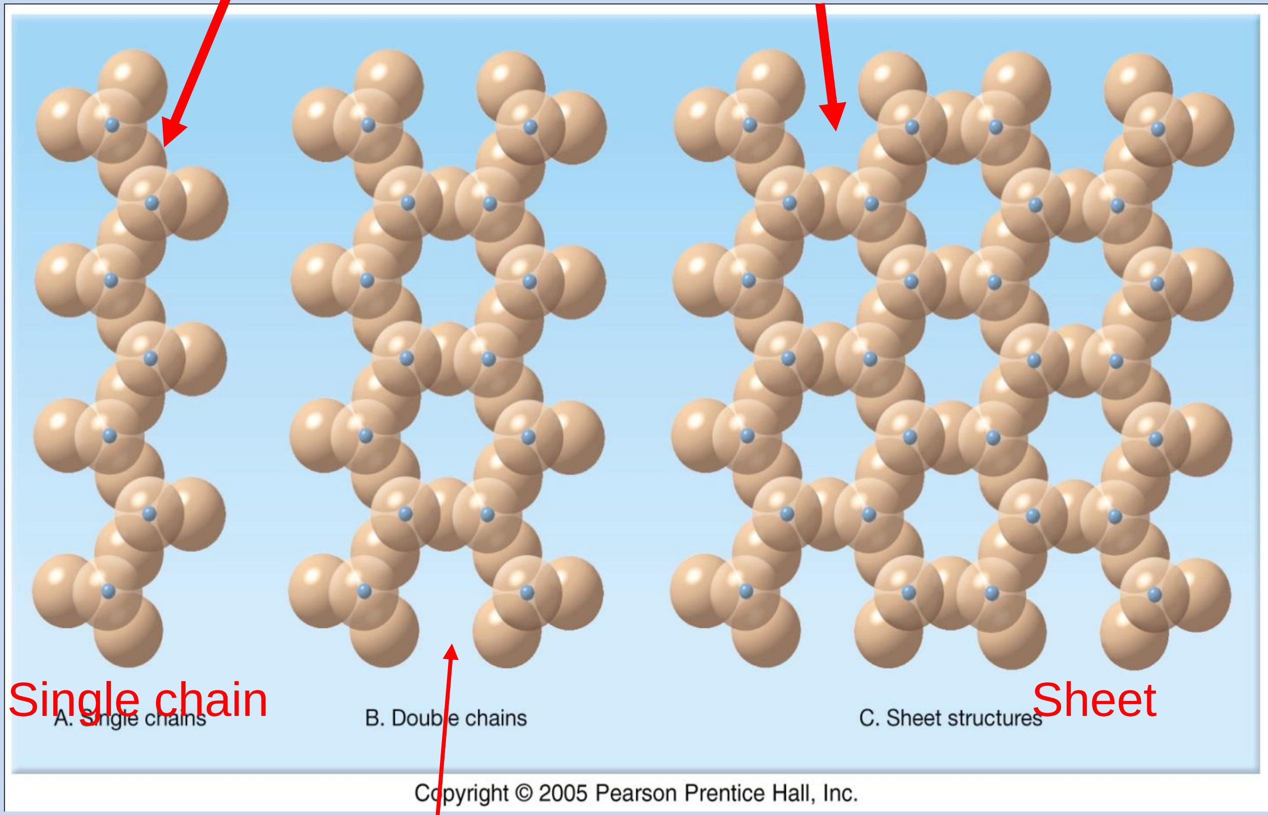
- Si-O tetrahedra link by sharing oxygens

Structural Types of Silicates

- Isolated tetrahedra
- Ring structures
- Single & double chains
- Sheet or layered
- Framework (3D)

2 oxygen ions linked to silicon

3 oxygen ions linked to silicons

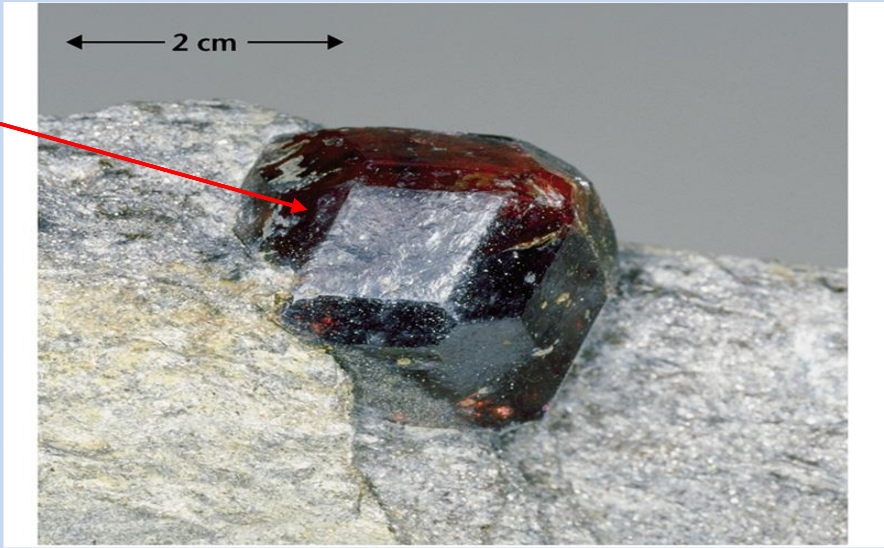


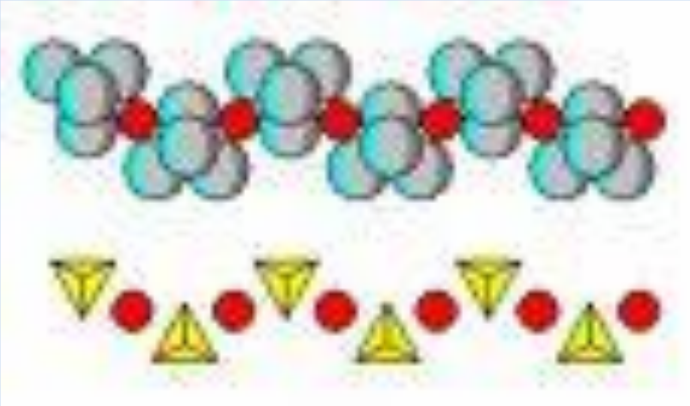
Double chain

Isolated Tetrahedra

- Tetrahedra not linked to each other

- Examples:
  - Olivine
  - Garnets



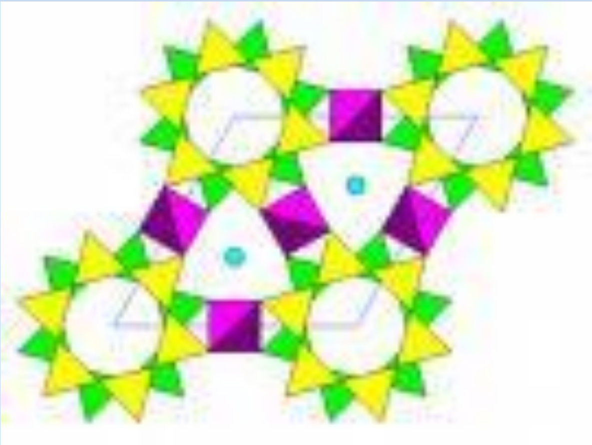


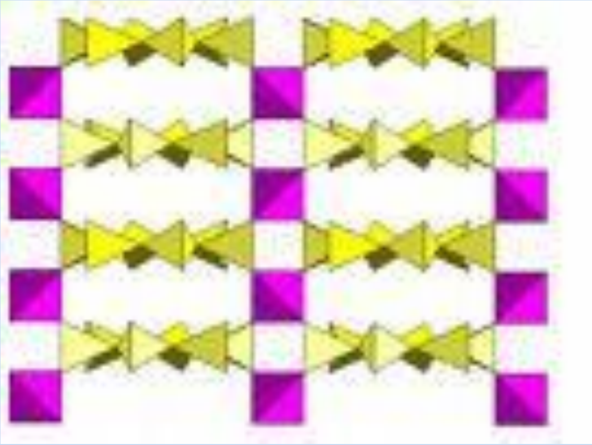
Ring Silicates

- Tetrahedra arranged in isolated rings

Examples:

- Beryl
- Tourmaline



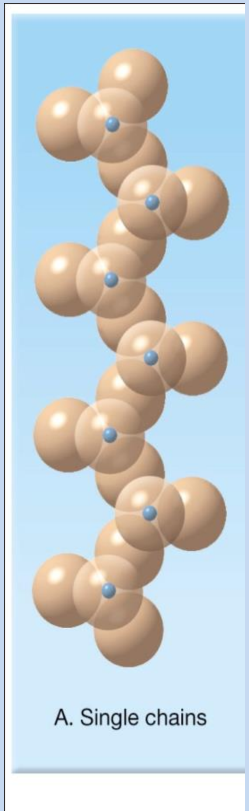


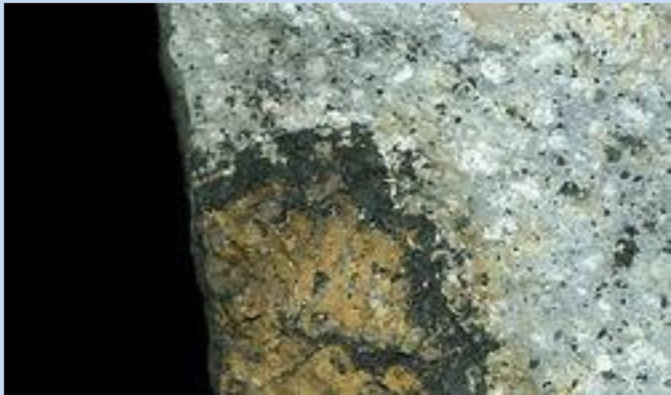
Single Chain Silicates

- Tetrahedra arranged in <u>single chain</u>

Example:

- – Pyroxenes

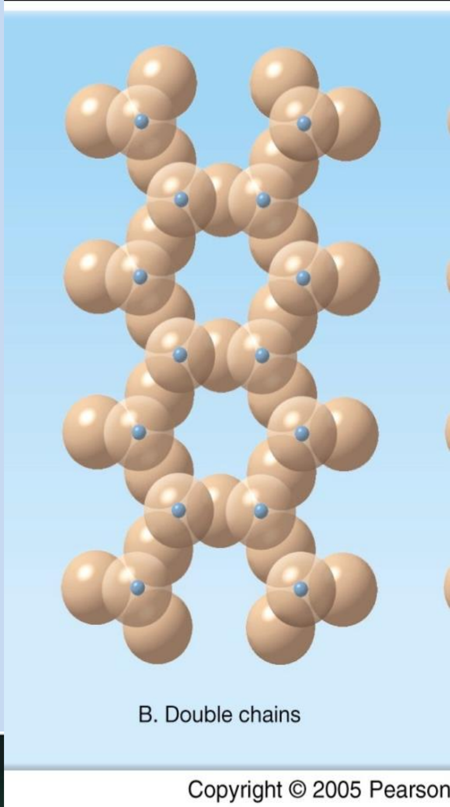




Double Chain Silicates

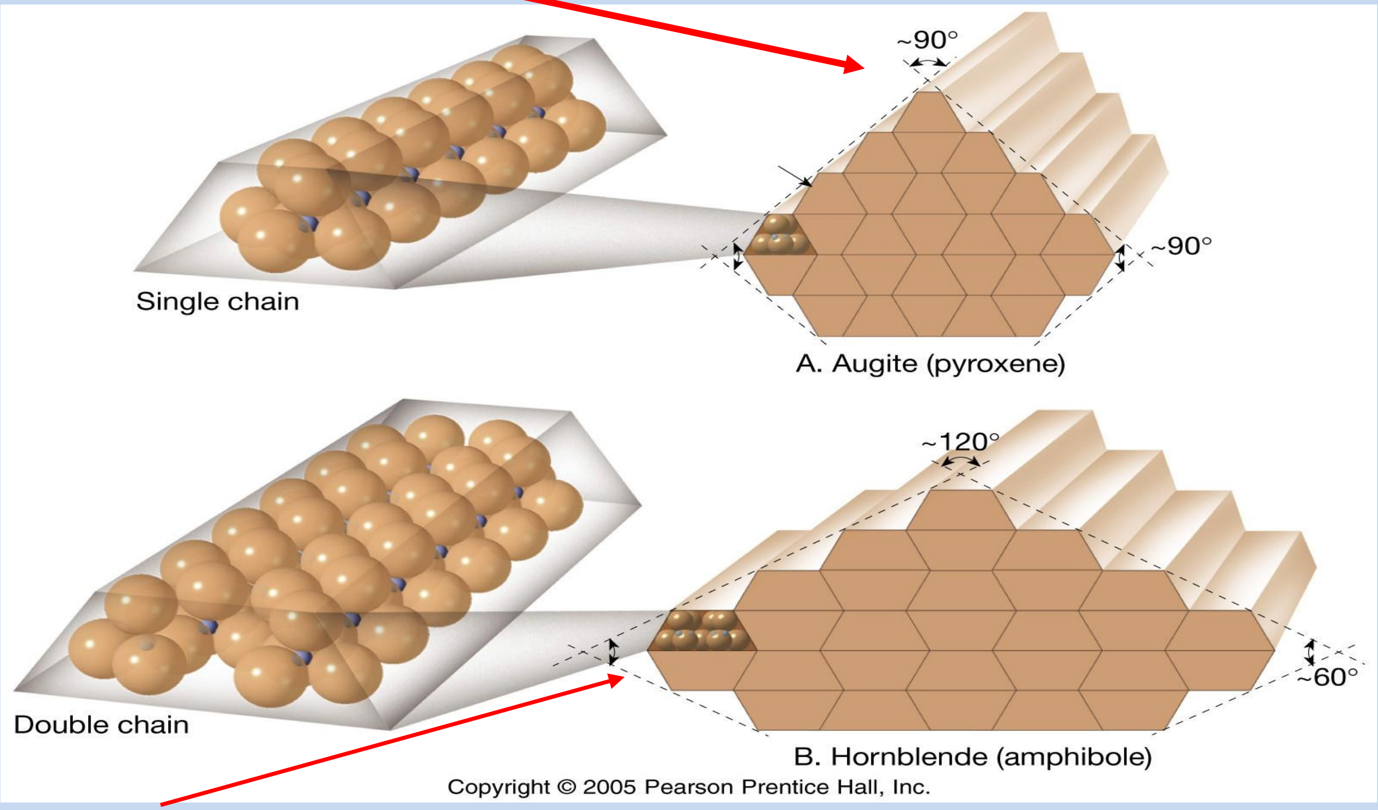
- Tetrahedra arranged in
  2 chains forming rings

Example:
- – Amphiboles



Cleavage Angles

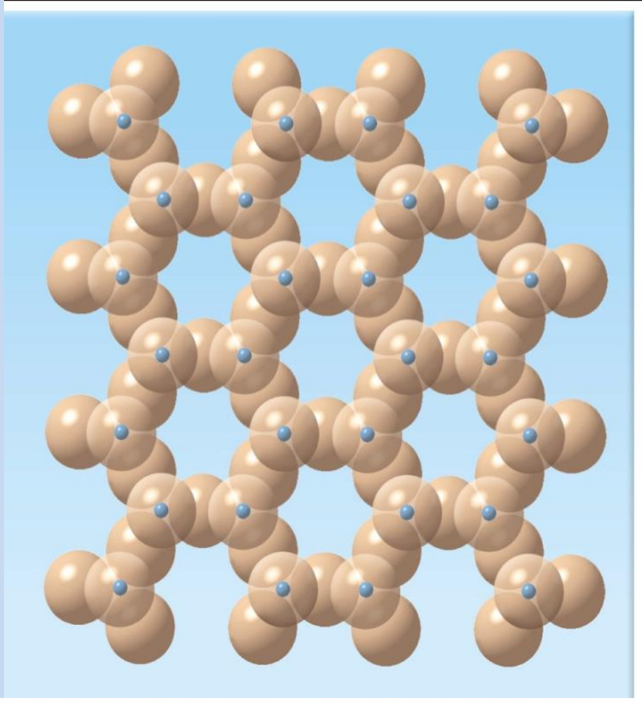
Pyroxene (Augite variety) cleaves parallel to the silicate chains - 2 at ~90° due to single-chain structure



Amphibole (Hornblende variety) has 2 cleavages at 60° & 120° due to double-chain structure

Sheet Silicates

- Si-O Tetrahedra form layered sheets held together by weak bonds

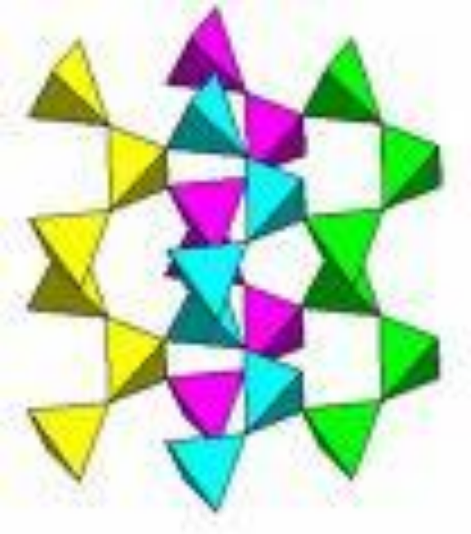


Examples:

- Micas
- Clay minerals

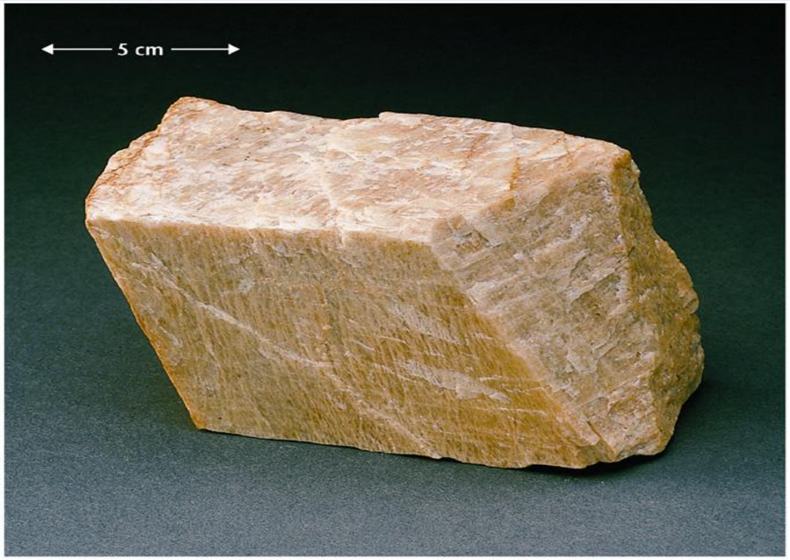
Framework Silicates

- Si-O tetrahedra form complex 3D structures



Examples:

- Quartz - SiO_2
- Feldspars



Minerals in Granite

5. Silicates are the most important mineral group